How the 2011 Room Reverb works

How the 2011 Room Reverb works (illustrated)

Polished companion to the text-first notes. Diagrams live in `diagrams/`.
Tags stay honest: **VERIFIED** = Structure / ens · **ASSUMED** / **PROPOSAL** / **from-them** = labeled as such.
This pack does **not** invent an A–F surface dictionary or a full Hadamard OCR sign table.

> Prefer formulas from `findings/structure-findings.md` and the engine locks in `software/engine/`. When audio disagrees with a guess, keep the VERIFIED algebra.

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1. What it is / design goals

The 2011 Reaktor ensemble models a rectangular room in two cooperating layers:

1. **Early field (image sources).** Mirror the source across walls for each reflection order, measure distance to a stereo mic pair, delay by distance ÷ speed of sound, and scale by geometric `1/r`. That supplies the first clear echoes and the L/R image of the room.
2. **Late field (Hadamard FDN).** A 32-line feedback delay network mixes energy with an **unscaled ±1 Hadamard** matrix and a single loop gain `g` in `(0, 1)`. Early taps excite the FDN; the early field itself stays a separate wet path into the mixer.

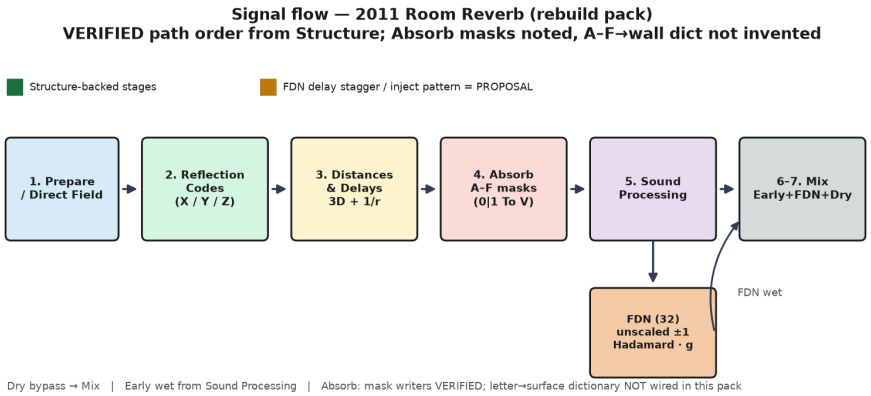
The Desktop rebuild copies **verified leaf algebra** and Prepare mic/source math into Python (`software/engine/early_field.py`, `fdn.py`, plus the tweak UI). It deliberately leaves unwired:

- A–F letter → physical wall dictionary
- A complete Structure OCR Hadamard sign table

Design goals for the pack: readable Structure locks, a slider UI that does not invent coefficients, and a clear VERIFIED vs ASSUMED boundary so A/B against the `.ens` stays honest.

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2. Signal flow



Signal flow

Order of work inside the instrument (and mirrored by the prototype):

Step	Block	What happens	Tag
1	Prepare / Direct Field	Panel knobs → metres; mic centre & source; Direct Field `DD` is 2D XZ	VERIFIED formulas
2	Reflection Codes X/Y/Z	Named mirror macros → image coordinates; To V collectors (~17–52)	VERIFIED macros
3	Distances and Delays	3D Euclidean `d`; `delay_ms = (d/340)×1000`; level `1/r`	VERIFIED
4	Absorb Diffus A–F	Serial codes write **0|1** masks into To V	Mask writers VERIFIED; wall dict not invented
5	Sound Processing	Per-voice delay, level, mic pattern, optional colour	Pattern LIKELY; filter

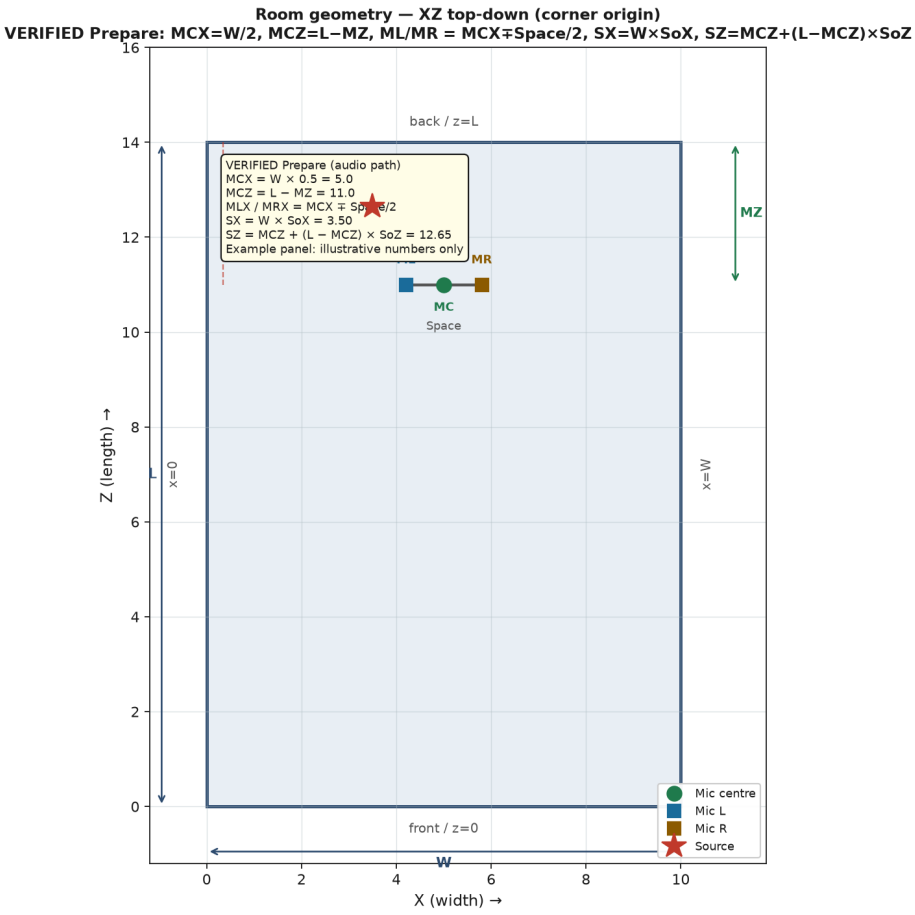
Step	Block	What happens	Tag
			coeffs ASSUMED
6	FDN	32 lines, unscaled ±1 Hadamard, ×g	Size/g lock from-them; matrix construction ASSUMED
7	Mix	Dry + early wet + FDN wet → out (`software/renders/out.wav`)	—

Dry audio bypasses the room path and joins at Mix. Early wet and FDN wet are separate sends.

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3. Room & Prepare geometry

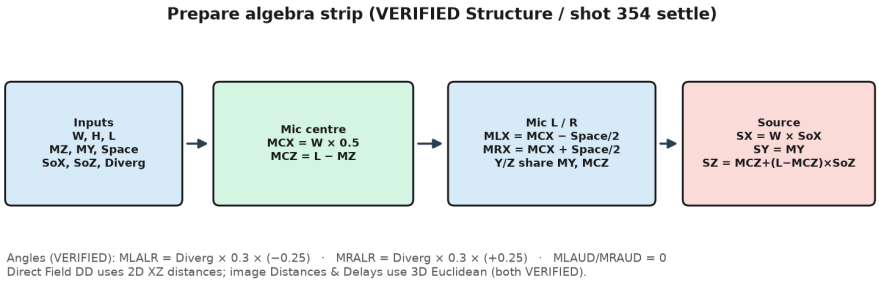
3.1 Top-down room (XZ)



Room geometry XZ

Corner origin: `(0,0)` at one corner, `W` along X, `L` along Z, `H` up Y (height is used in 3D image distances and Y-mirrors, not in this plan view).

3.2 Prepare algebra strip



Prepare algebra

Symbol	Formula	Tag
MCX	`W × 0.5`	VERIFIED
MCZ	`L − MZ`	VERIFIED / LOCKED (settled vs earlier `L+MZ` / `L×MZ` misreads)
MLX / MRX	`MCX − Space/2` / `MCX + Space/2`	VERIFIED
Mic Y / Z	`MLY=MCY=MRY=MY; MLZ=MCZ=MRZ=MCZ`	VERIFIED
MLALR / MRALR	`Diverg × 0.3 × (−0.25)` / `× 0.3 × (+0.25)`	VERIFIED
MLAUD / MRAUD	`0`	VERIFIED
SX	`W × SoX` (SoX in 0–1)	VERIFIED
SY	`MY`	VERIFIED
SZ	`MCZ + (L − MCZ) × SoZ`	VERIFIED

Panel Mic Stand drawing math is a **separate** display path. Soft OCR on that path must not override these Prepare audio locks.

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4. Image-source early field + verified leaf table

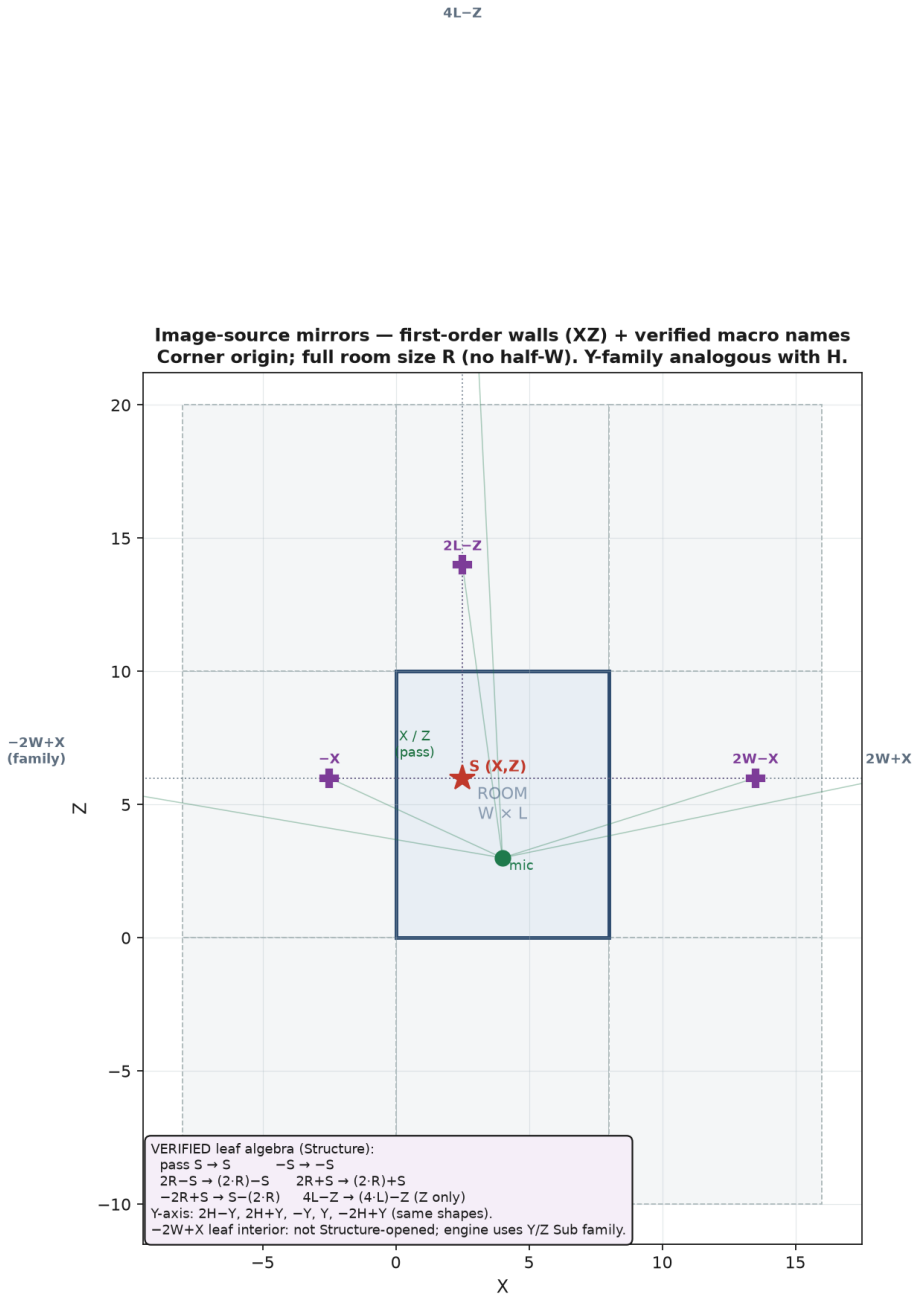


Image-source mirrors

Reflection Codes build image coordinates per axis from named macros, then Distances and Delays turn each image into time and level.

4.1 Verified leaf algebra

Corner origin, full room size `R` (no half-`W`). Sub pin order as opened in Structure.

Macro / law	Formula	Tag
Pass-through X / Y / Z	S	VERIFIED
−X / −Y	−S	VERIFIED (−X shot 353; −Y 319)
2W−X / 2H−Y / 2L−Z	(2·R) − S	VERIFIED
2W+X / 2H+Y / 2L+Z	(2·R) + S	VERIFIED
−2H+Y / −2L+Z	S − (2·R)	VERIFIED
−2W+X	same Sub family as Y/Z	leaf not opened; engine uses same shape
4L−Z	(4·L) − Z	VERIFIED (Z only)
Speed of sound c	340 m/s	VERIFIED Const
Delay	(d/340)×1000 ms	VERIFIED
Geometric level	1/r	VERIFIED
Image distance	3D Euclidean	VERIFIED
Direct Field DD	2D XZ	VERIFIED

4.2 Prototype image grid

32 images = 3×3×3 plus face centres, omit (0,0,−2) (**from-them** / agreed). Polarity (−1)^(nx+ny+nz) (**from-them**).

To V collectors share indices across X/Y/Z Codes so one index is one (X,Y,Z) image voice. X Code maps macros onto repeating indices **17–52** (batch-4). Y/Z fan-outs are partially documented in findings; the full index→macro table is still incomplete for OCR soft rows.

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5. Absorb A–F masks (known vs not invented)

Structure shows a serial **A Code → … → F Code** chain. Each stage writes **0 or 1** into indexed To V slots (same family of collectors as Reflection Codes). That is a **surface → voice mask** bus.

What	Status
A–F mask writers exist and feed To V	VERIFIED
Bit tables OCR’d in batch-4 notes	Present in findings — **not** claimed as audio-engine truth here
Letter A–F → floor / ceiling / wall dictionary	**NOT invented** in this pack
Panel Surfaces / Diffusion Matrix On/Off gates	VERIFIED UI; LIKELY tied to A–F — still not a wired dictionary in the prototype

The rebuild therefore does **not** mute images by wall letter. Treat long-tail absorb colour differences vs Reaktor as expected until that map is Structure-proven and intentionally wired.

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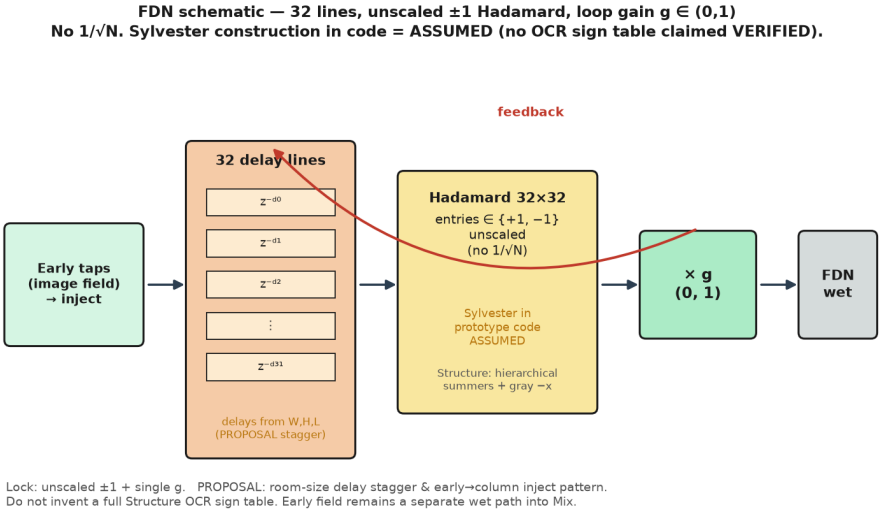
6. Mic model

Stereo capsules sit on a shared stand:

- Positions from Prepare (section 3).
- Toe angles `MLALR` / `MRALR` from Diverg (VERIFIED product; not raw degrees).
- Pattern: panel **Car/Omni** blends toward cardioid. Prototype keeps a **LIKELY** `(1+cosθ)/2` lerp; exact Core `0x7e` algebra is unproven.
- `Dir+` port comment says radians; Primary trig units elsewhere remain turns — verify before trusting sine units.
- `Res` / Mic Response is an HF limit (Hz-scale on panel); do not invent dB/Q from soft OCR.
- Humidity and peaking colour in the prototype are **ASSUMED** gentle shelves / Q — tagged in engine comments.

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7. Unscaled Hadamard FDN



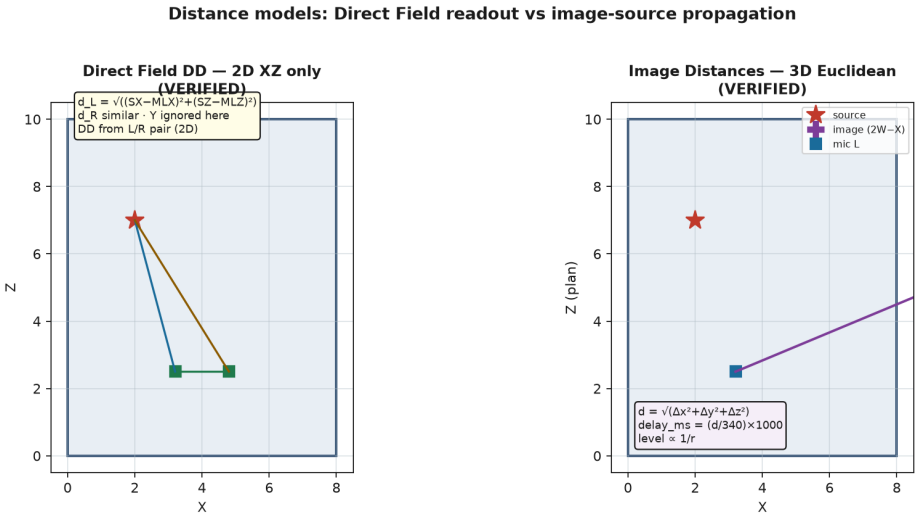
FDN schematic

Rule	Value	Tag
Size	N = 32	agreed / from-them
Matrix	unscaled ±1 Hadamard	construction ASSUMED (Sylvester in code); no OCR rows claimed VERIFIED
Loop gain	`g ∈ (0, 1)`	from-them / lock
Scaling	**no `1/√N`**	lock
Delays	stagger from room `W,H,L`	PROPOSAL
Inject	early taps → Hadamard columns	PROPOSAL pattern

Structure shows hierarchical summers, gray `−x` for signs, and in one shot an `In×(1/32)×Gain` path. The live audio FDN copy is not fully restored; this prototype keeps the agreed **unscaled ±1 + `g`** rule and refuses to paste soft OCR rows as verified coefficients.

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8. Direct Field 2D vs image 3D



Direct vs image distance

Both laws are VERIFIED and must not be collapsed into one:

Path	Distance	Use
Direct Field	2D XZ only	`DD` readout / direct path UI
Image Distances and Delays	3D `√(Δx²+Δy²+Δz²)`	early taps: delay + `1/r`

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9. Verified vs assumed summary

Locked (prefer these in A/B)

- `c = 340`
- `MCZ = L − MZ`, `MCX = W × 0.5`, ML/MR Space split, source SX/SZ formulas
- Mirror leaf algebra in the table above
- Unscaled ±1 FDN, no `1/√N`, `g ∈ (0,1)`
- No A–F surface dictionary wired

Still ASSUMED / PROPOSAL / open

- Sample rate 44100
- Peaking Q, humidity shelf coeffs, allpass milliseconds
- Cardioid exact `0x7e` blend
- FDN delay stagger and inject mapping
- Motion hop/OLA details in the moving-source demo

- Full Hadamard sign table from Structure OCR
- A–F → wall dictionary
- `−2W+X` leaf interior (family assumed from Y/Z)

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10. How to A/B with Reaktor + tweak app

1. Keep the ensemble on your machine (Desktop pack `ensemble/` note; you already have `2011-room-reverb.ens` / `20 October 2011 room reverb.ens`).
2. Open **`software/Launch Tweak App.bat`** (or `launch_tweak_app.sh` on the box).
3. Feed the **same dry WAV** into Reaktor and into the Tweak App.
4. Match room / mic / source knobs as far as labels allow: `W`,`H`,`L`, Space, Diverg, SoX/SoZ, Car/Omni, GainF-style `g`.
5. Render with **path motion off** first for a static A/B; then try ellipse motion vs a fixed Reaktor source.
6. Compare early clarity and L/R image first. Treat long FDN tail differences as expected while delay stagger remains PROPOSAL and A–F masks are unwired.
7. Write output to `software/renders/out.wav`. Desktop demo WAVs are listening references only — not bit-exact Reaktor claims.

When something disagrees, prefer Structure **VERIFIED** algebra over prototype **ASSUMED** filters. Do not “fix” disagreement by inventing coefficients.

Optional engine check from `software/`:

```
.venv\Scripts\activate
cd engine
python selfcheck.py
```

Further reading

Doc	Role
HOW_IT_WORKS.md	Text-first twin (no diagrams)
QUICKSTART.md	Launch / render steps
../findings/structure-findings.md	Master Structure log
../findings/INDEX.md	Batch note index
../software/README.txt	Tweak app entry

Doc	Role
`/workspace/panel-controls.md`	Panel tab catalog (batch-6)

Diagram sources: regenerate with `python docs/generate_diagrams.py` from the pack root (needs matplotlib in `software/.venv`).